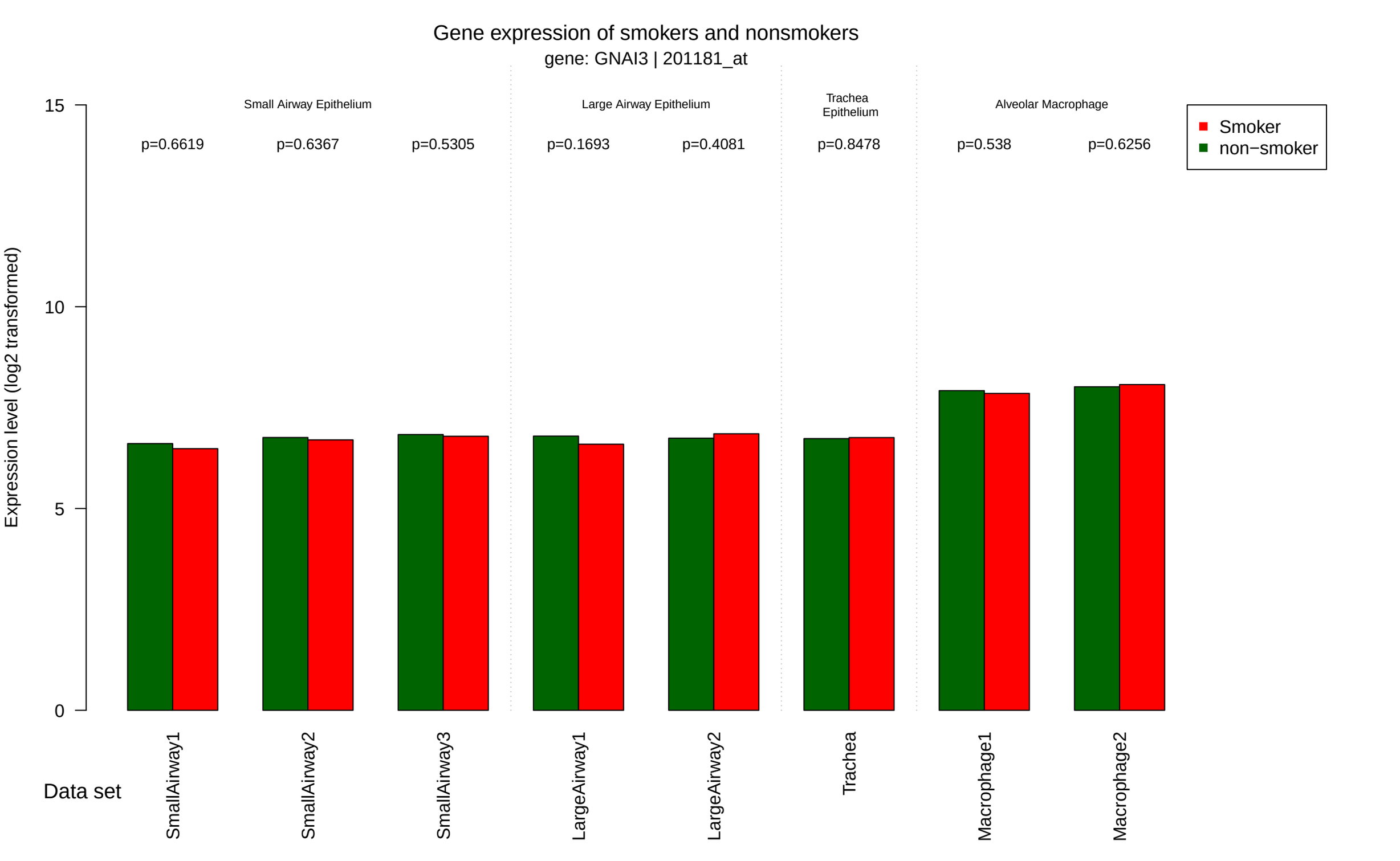
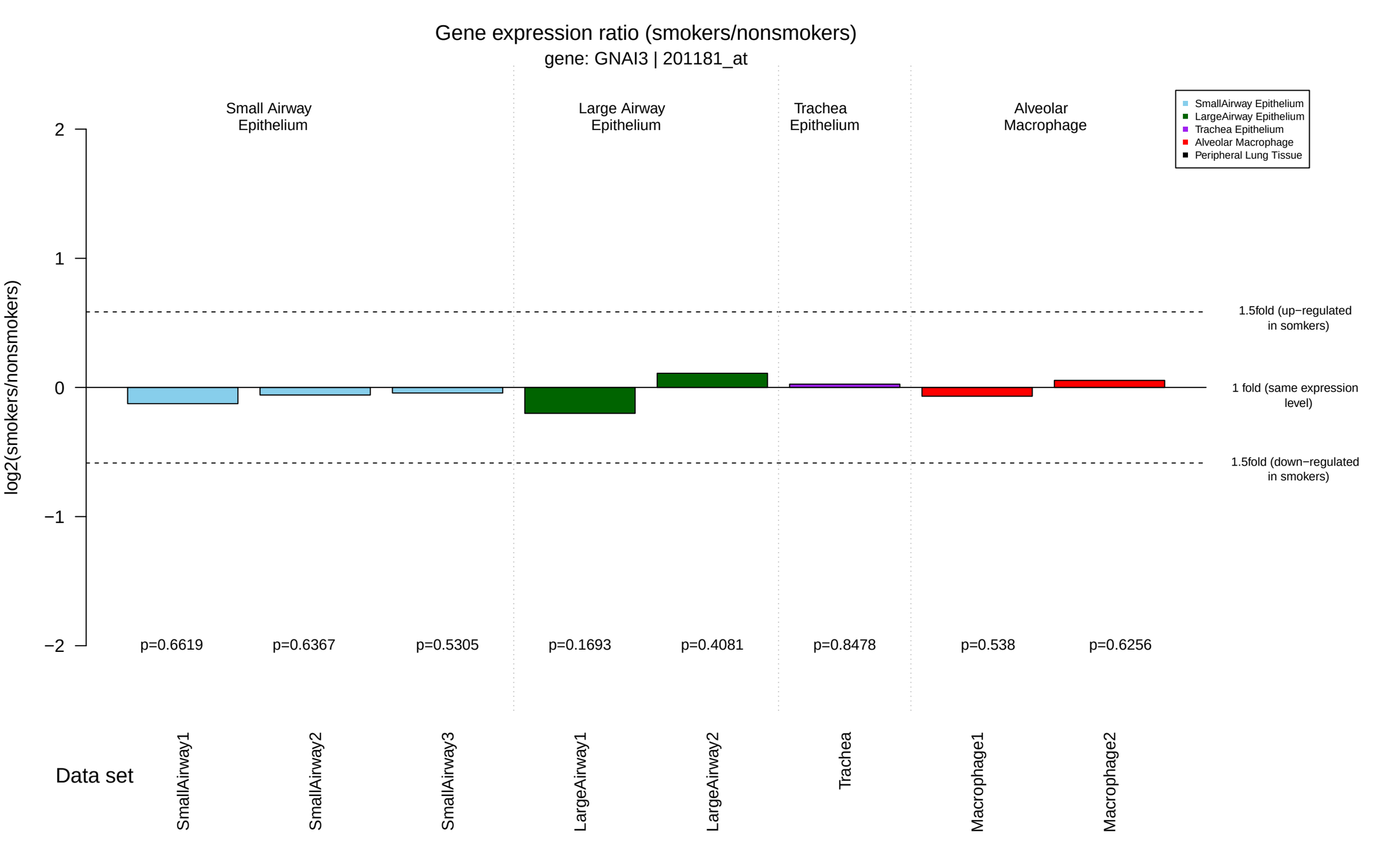
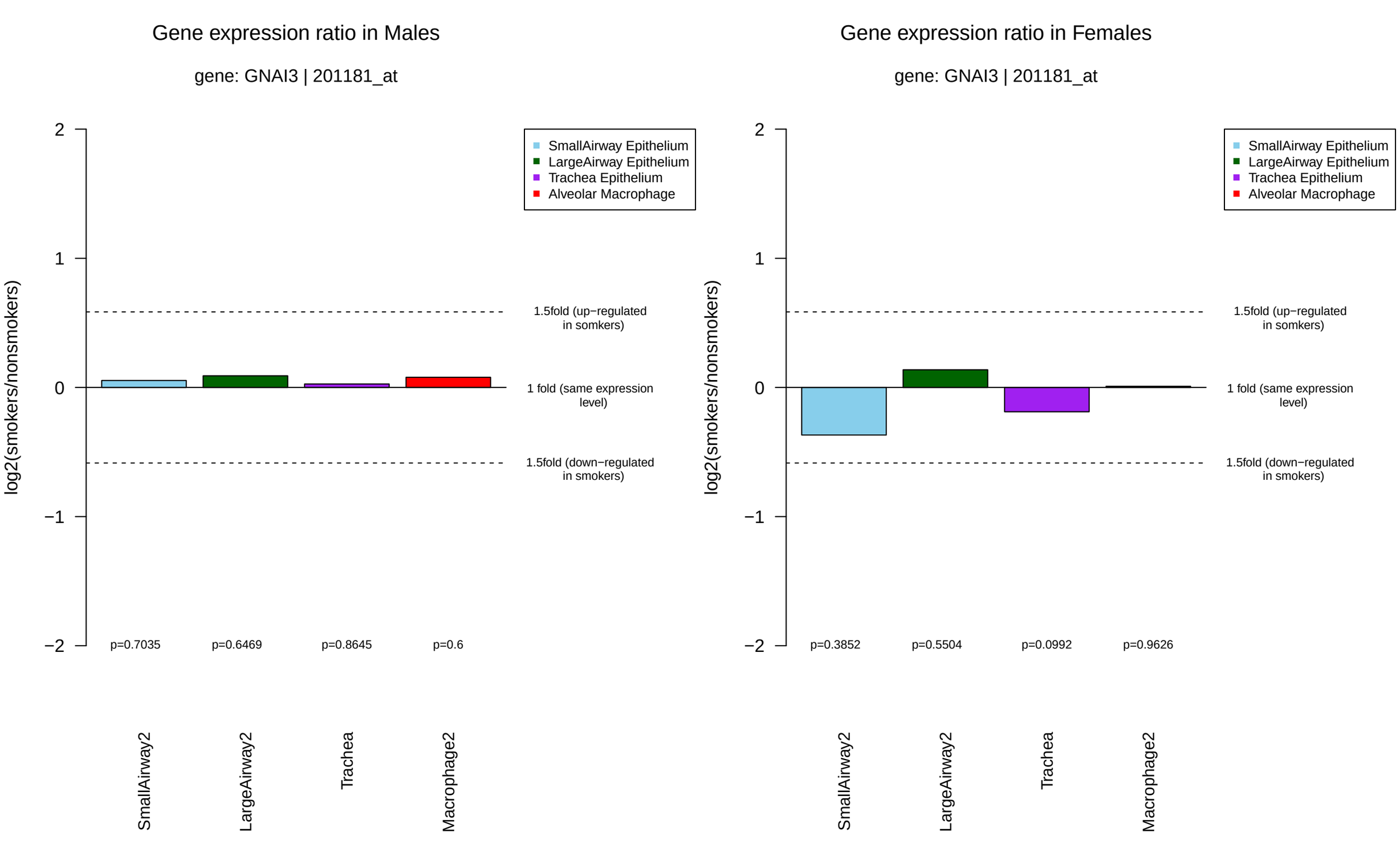
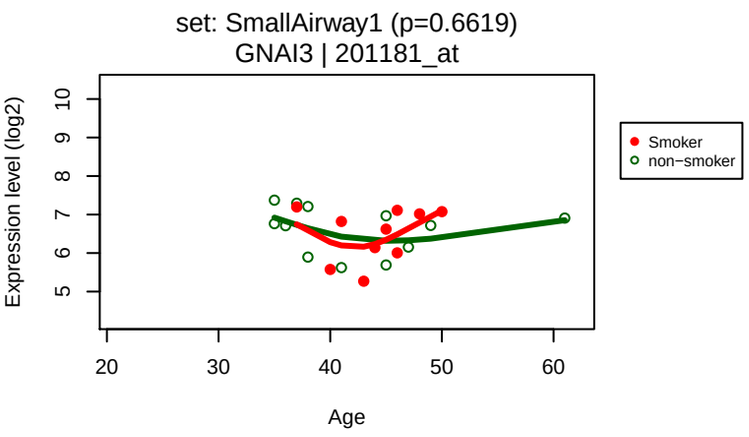
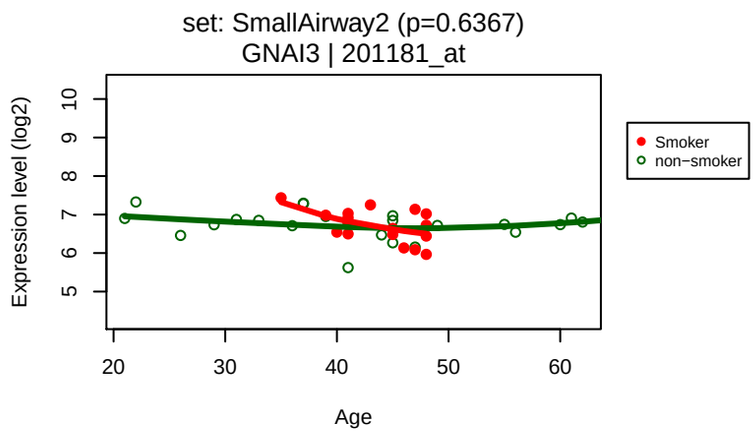
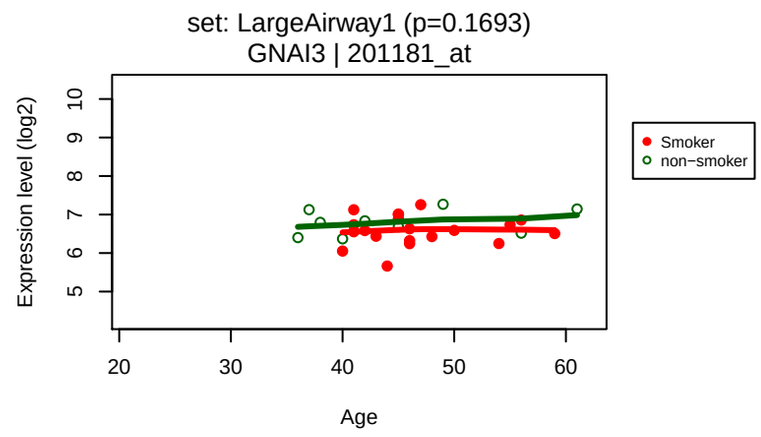

Gene expression of smokers and nonsmokers
gene: GNAI3 | 201181_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6619
p=0.6367
p=0.5305
p=0.1693
p=0.4081
p=0.8478
p=0.538
p=0.6256
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

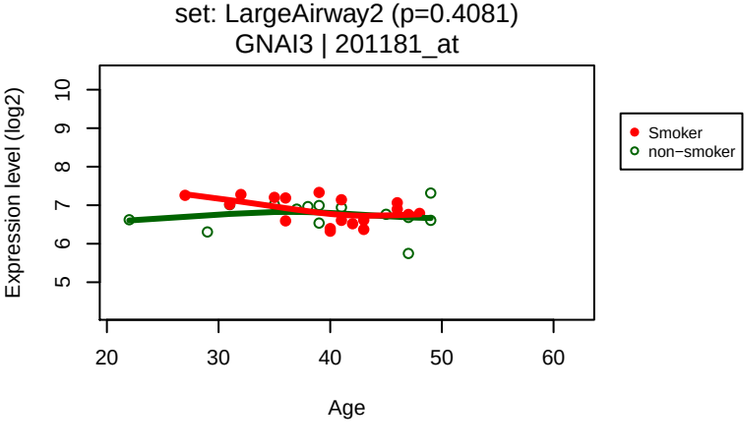

Gene expression ratio (smokers/nonsmokers)
gene: GNAI3 | 201181_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6619
p=0.6367
p=0.5305
p=0.1693
p=0.4081
p=0.8478
p=0.538
p=0.6256
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

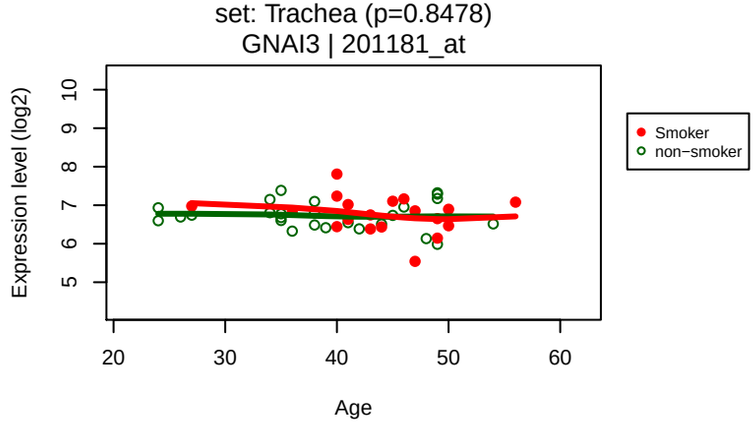

Gene expression ratio in Males
gene: GNAI3 | 201181_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7035
p=0.6469
p=0.8645
p=0.6
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAI3 | 201181_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3852
p=0.5504
p=0.0992
p=0.9626
SmallAirway2
LargeAirway2
Trachea
Macrophage2

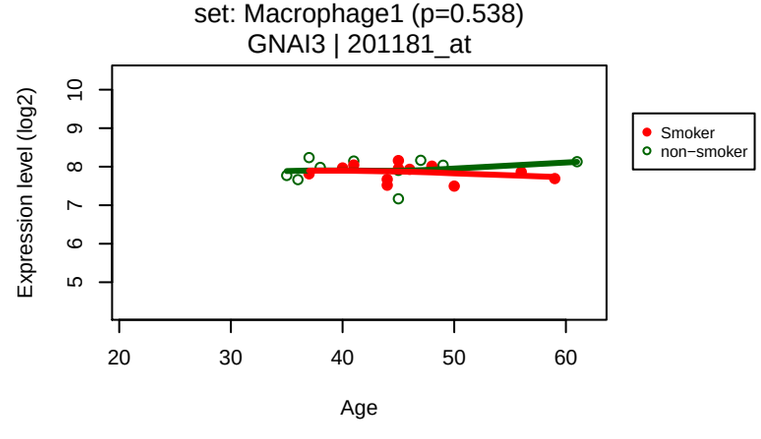

set: SmallAirway1 (p=0.6619)
GNAI3 | 201181_at
Expression level (log2)
Age
Smoker
non−smoker

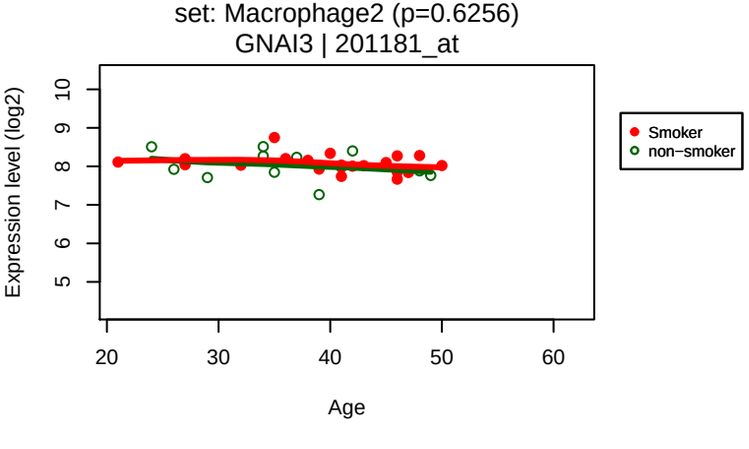

set: SmallAirway2 (p=0.6367)
GNAI3 | 201181_at
Expression level (log2)
Age
Smoker
non−smoker

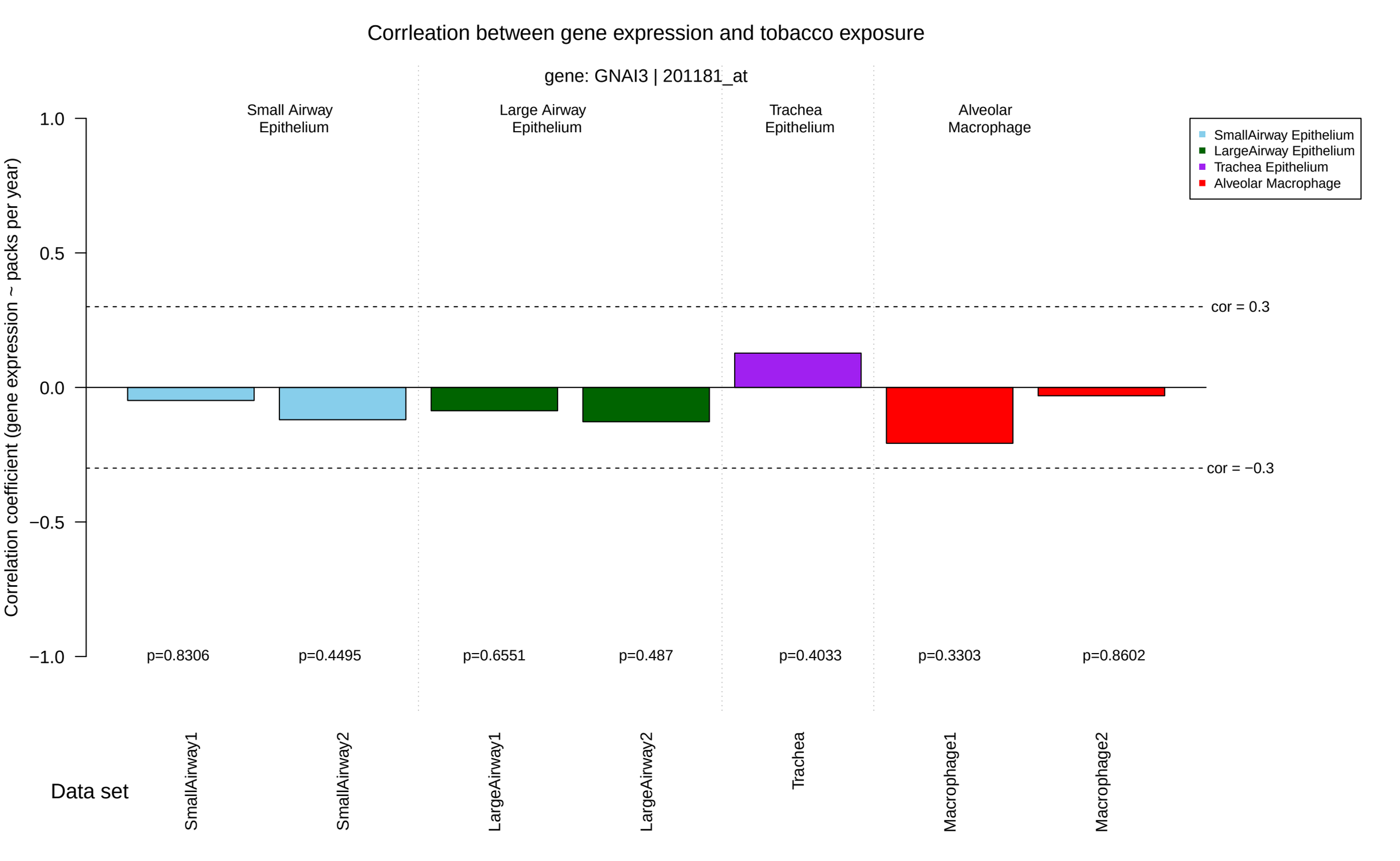

set: LargeAirway1 (p=0.1693)
GNAI3 | 201181_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4081)
GNAI3 | 201181_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8478)
GNAI3 | 201181_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.538)
GNAI3 | 201181_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6256)
GNAI3 | 201181_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAI3 | 201181_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8306
p=0.4495
p=0.6551
p=0.487
p=0.4033
p=0.3303
p=0.8602
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2